CUSTOMER SEGMENTATION AND CHURN PREDICTION IN TELECOMMUNICATIONS COMPANIES

Combining clustering, classification, and synthetic tabular data to understand churn risk in telecom customers

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NUS National University of Singapore

2,666 TRAINING SET

667 HOLD-OUT TEST SET

14.6% BASELINE CHURN RATE - HEAVY CLASS IMBALANCE

12 RETAINED PREDICTORS IN FEATURE SPACE

UNDERSTAND
The Customer and the Problem

PREDICT
The Machine Learning Engine

EXTEND & ACT
Gen-AI Stress Test and Business Value

PROBLEM & OBJECTIVES

Why it matters

Customer segmentation and churn prediction are two critical data mining tasks in telecommunications, where retaining an existing customer is often less costly than acquiring a new one. This project studies churn from unsupervised, supervised, and Gen-AI perspectives.

Objectives

- **Unsupervised learning:** identify meaningful customer segments
- **Supervised learning:** predict customer churn from historical data
- **Gen-AI extension:** evaluate synthetic structured data for downstream analysis

DATASET & PREPROCESSING

The dataset contains 3333 customer records (2666 training / 667 hold-out test) and 20 columns covering account attributes, service plans, usage metrics, and customer-service interaction.

Preprocessing

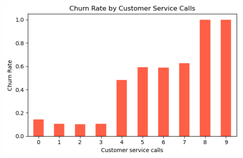
- No missing values were detected in the raw data.
- Categorical variables encoded: Churn, International plan, and Voice mail plan.
- Train-only standardization was applied to retained numerical features.
- Removed redundant or weak-signal columns:

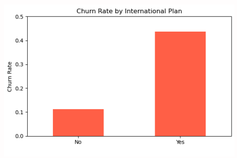
Dropped columns

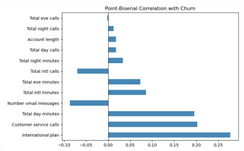
- 4 charge columns
- Voice mail plan
- Area code
- State

EDA — KEY CHURN DRIVERS

International plan and repeated customer-service calls are the clearest churn signals in training data.



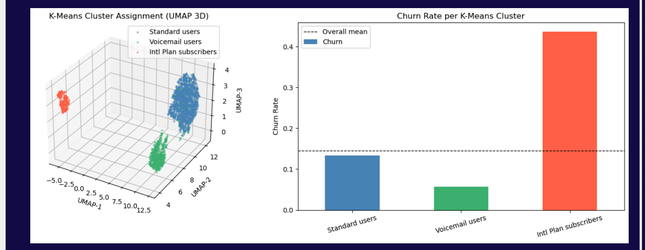




Point-biserial analysis identifies **International plan, Customer service calls, and Total day minutes** as the strongest churn-associated variables. The dataset is also class-imbalanced, meaning raw accuracy alone is not sufficient for model evaluation. Churn rises sharply after repeated support contact, suggesting a threshold effect rather than a smooth linear increase in risk.

CUSTOMER SEGMENTATION (K=3)

K-Means reveals three interpretable customer groups with sharply different churn risks.



Why K=3?
Although K=2 remained the stronger metric-based baseline, K=3 was selected because it produced a more useful business segmentation by separating the customer base into standard users, voicemail users, and international-plan users, each with clearly different churn risk.

DBSCAN insight: DBSCAN did not recover a stable 3-cluster structure, suggesting that the customer base is more centroid-separable than density-separable.

SUPERVISED LEARNING LEADERBOARD

Random Forest is the best overall model on the held-out real test set.

Random Forest (Best Overall)

- F1: 0.804
- ROC-AUC: 0.926
- PR-AUC: 0.869
- Accuracy: 0.945

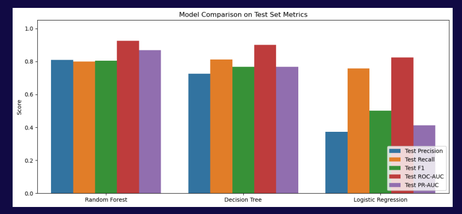
Decision Tree (Recall-Oriented)

- Highest Recall: 0.811
- Useful when missing a churner is costly.

Logistic Regression (Linear Baseline)

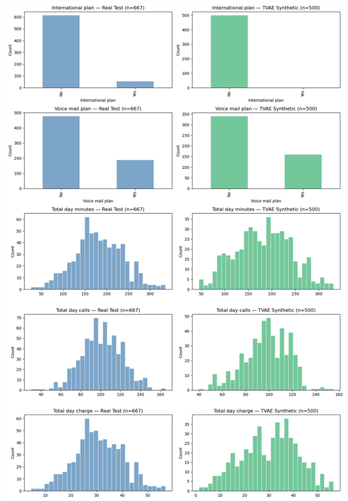
- F1: 0.500
- Weaker fit on this dataset.

Three classifiers were tuned using training-only cross-validation, with F1-score treated as the main selection criterion because of class imbalance. Random Forest provides the strongest overall balance across F1, ROC-AUC, PR-AUC, and accuracy.



GEN-AI DIAGNOSTIC (REAL VS. TVAE SYNTHETIC)

TVAE broadly preserves several statistical patterns, but fails to fully preserve the rare conditional structure needed for stable predictive fidelity.



Why TVAE was retained?

CTGAN, TVAE, and Gaussian Copula were explored as candidate tabular synthesizers. TVAE was retained for detailed follow-up because it preserved churn balance and supported downstream stress-testing, even though some marginals were matched more closely by other methods.

What failed most clearly?

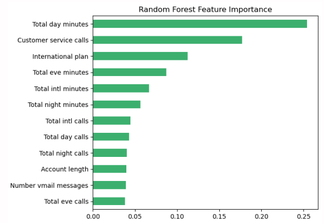
Intl Plan = Yes: 7.9% → 0.2%
Random Forest F1: 0.804 → 0.746
The main failure lies in underproducing rare but high-impact categories.

Method Selection Summary

- CTGAN: flexible synthesis, but less stable on churn balance
- TVAE: best overall tradeoff for follow-up analysis
- Gaussian Copula: matched some marginals better, but weaker for downstream stress-testing

FEATURE IMPORTANCE

The same signals recur across EDA and supervised learning: International plan remains the clearest high-risk indicator, Customer service calls captures escalating service friction, and Total day minutes provides a strong usage-intensity signal, making them the most informative churn features.



KEY LIMITATION OF SYNTHETIC DATA

Synthetic data is useful for stress-testing pipelines, but it is not reliable enough to replace real-data evaluation when rare, high-impact categories matter. In this project, TVAE preserves some broad marginals while smoothing away rare churn-relevant structure, especially for International plan = Yes. This shows that matching overall distributions is easier than preserving the exact customer-risk structure needed for stable prediction.

ACTIONABLE RETENTION STRATEGIES

TARGET: Prioritize retention monitoring for international-plan customers, the clearest high-risk segment.

ESCALATE: Trigger proactive service recovery after repeated customer-service calls, before dissatisfaction hardens into churn.

ANALYZE: Use voicemail-heavy users as a low-risk benchmark to identify features associated with customer stickiness.

CONCLUSION

Churn in this telecom dataset is concentrated in identifiable customer profiles rather than being uniformly distributed across all users. K-Means reveals actionable customer segments, and Random Forest provides the strongest overall predictive performance. Overall, the project shows how data mining can move from descriptive churn signals to segmentation, prediction, and practical retention intervention.

LESSONS LEARNED

- Interpretability matters: business-useful customer segments may be more valuable than metric-optimal clustering alone.
- Prediction is not enough: strong churn models are most useful when paired with interpretable risk signals and customer profiles.
- Synthetic data has limits: TVAE is useful for stress-testing, but cannot replace real evaluation when rare, high-impact categories matter.

TEAM CONTRIBUTIONS

- Lin Chieh — Part 1: Data Preprocessing; Part 2: Exploratory Data Analysis (EDA)
- Liu Fan — Part 3: Unsupervised Learning – Customer Segmentation
- Liu Zimeng — Part 4: Supervised Learning – Churn Prediction
- Zhang Yizhuo — Part 5: Generative AI to Enrich the Training Dataset
- All team members — Poster design, integration, and final presentation materials